Agenda: How to run Custom...

problem statement:

Text2SQL APP → TBS (Tokyo Broadcasting System Television).

↙

Domestic Broadcast right.

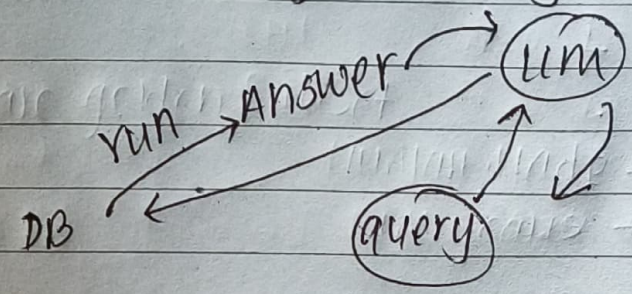
↙

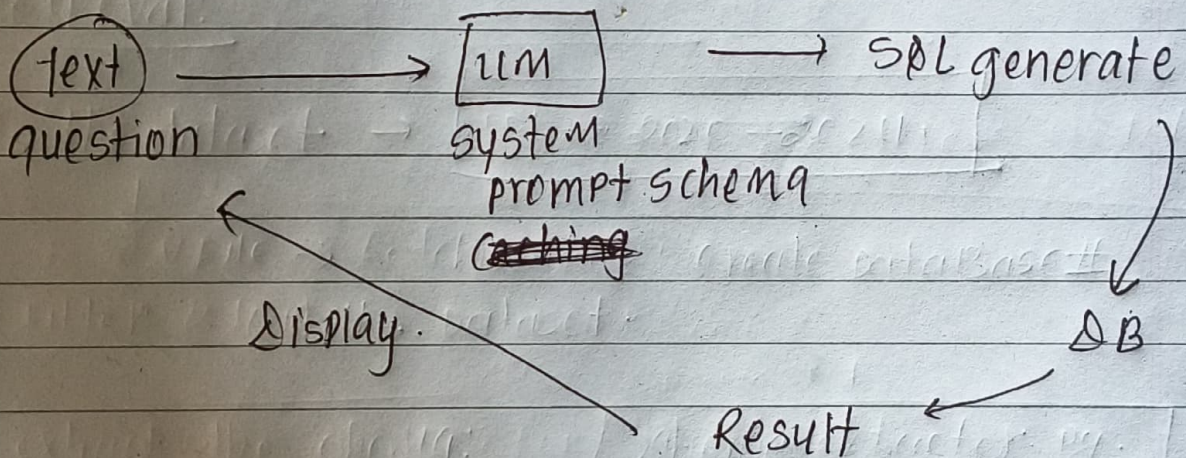
TBS → platform streaming

↓

Add a feature Text2SQL at scale.

goal select suitable Best Model For Text2SQL with in Budget ceiling.





Think??
Deeply

Why can't we use Best Models like Fable?

what about custom evals?

what about directly selecting from leaderboard?

LLM LeaderBoard Job is not to select Model.

it is used to Filter Models Based upon our requirement.

Select 5-10 models from leaderboar LLM stats & then run custom evals.

Flow

We use the LLM leaderboard to Filter

1) your requirements.
2) shortlist 5-10 models.
3) custom model evals.

Requirements:

1. task → Text2SQL
2. cost → Retention play → they want us to spend more time on website.
   ↓
   cost ceiling → 3 lakhs.

Text2 SQL system.

[LLM] → Per token cost.

Text → [application] → SQL query.

↓

prompt

system:

You are a text-to-SQL generator. given a database schema return a single SQL query.

DB schema + query?

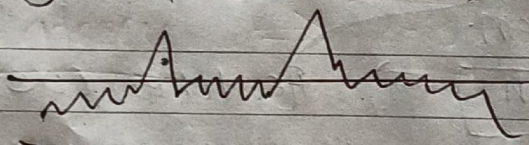
LLM charges for: input tokens + output tokens.

Calculate input tokens:
use token calculator.net.

≈ 352 tokens. ≅ 400 tokens.

Calculate output tokens:
use tokencalculator.net
≅ 100 tokens.

on avg daily → questions ⇒ 30K



Aug ≈ 50K

19 → 400 input | 100 output:

→ 50000 x 30 days x 95 usd to ~~₹~~ inR.

FableCost:

Claude Fables	input	output:
	10 $/M	50 $/M

Per query cost: $\frac{10}{1000000} \times 400 + \frac{50}{1000000} \times 100$

$\frac{4}{1000} + \frac{5}{1000} = 0.009$

= 0.009 usd/query.

perday

=50K = 0.009 x 50K = 450 $ daily

permonth

450$/daily * 30 = ~~1800~~ 13500 $/Month.

13500 $/month * 95 (Conversion rate) = 1282500

Rs

But our Budget is only 3lakh/month. ≈ 4 times More Budget Needed.

Check Models of claude who have 1/4th the input & output token cost.

Refresh cycle = 5min or 1hr.

Imp discussion

5m cache writes.	1h cache writes.	cache Hits & Refreshes
$12.50/MTOK	$20/MTOK	$1/MTOK.

prompt Caching

⇓

To save the input TOKEN.

Significantly Reduce in Cost:

	Base input Tokens.	5M cache corites	1hr cache corite	Cache Hits & Refreshes
Claude Fables.	$10/M TOK	$12.5/MTOK	$20/MTOK	$1/MTOK.

you should know about all of above terms.

How caching works:

KV cache

Attention mechanism

12 lakh to nearly 6 lakh.

3rd point

Latency ⇒ 2-3 sec. ⇒ if not OCD Triggers:

4th point:

Context

↳ context coindow doesn't matter.

5th point

Deployment

↳ you can use the API keys.

6th point

Correctness

↳ Correctness Matters the most.

step 1: 1st goal write Requirement:

gather requirement:

1. Task
2. Cost
3. latency
4. Context Window.
5. Deployment
6. correctness.

step 2 leaderboard (5-10) Models filter:

check text to sql leaderboards → Not updated.
→ Don't Fully Trust ~~your~~ online leaderboards.

go to the coding leaderboard as aliace.

llmstats.com they have good information.

Kimi K3

How to get the Blended number

$$\frac{10X4 + 1X50}{5} = \frac{90}{5} = \$1.8$$

1st step to choose models From LeaderBoard:

Cost ceiling 3$/Month.

step1 Calculate Cost per Month. For Models & eliminate Models who will cost us more than 3$/Month.

step2 Do Normalization of Rating For all the remaining Models.

Val

~~So min~~
~~So max~~.

Normalize it [0 to 7].

step3 Normalize the speed:

$$\frac{Val - min}{max - min}$$

Normalize

step4

0.9 * normalized rating score + 0.1 * normalized latency score.

90% weight to Coding Capability + 10% weightage to latency score

Why 10% weightage to latency

SQL query hardly 100 to 200 rows.

So to write 100 to 200 rows very easy for every model

Now we will get our top 10 models:

	Model	Index	RS/40	Throughput	Score
1.	GPT-5.6-Tera	46.0	4.47L	57 c/s	0.909
2.	Kimi K3	44.9	4.834	4 c/s	0.883
3.	Muse Spark 1.1	38.3	1.654	266 c/s	0.823
4.	Minimax M3	35.4	42K	171 c/s	0.76

Qwen
GLM

→ Open source models

Do Custom evals on these models:

Test
1. GPT 5.6 Tera
2. Kimi K3
3. Muse Spark 1.1
4. Grok 4.5
5. Claude Sonnet 5
6. Minimax M3.

step 1: gather requirements.

step 2: shortlist 5-10 models from leaderboard.

step 3: Custom evals:

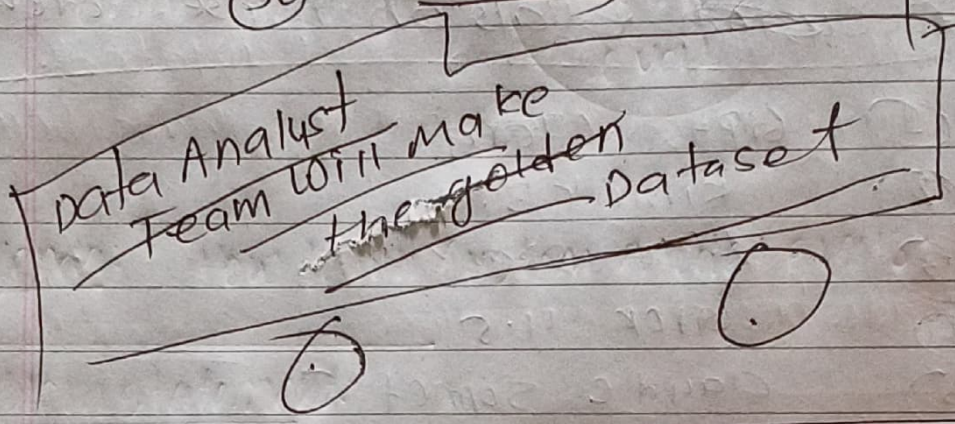
① Dataset: → Kaggle IPl dataset

② we need golden dataset.

30-50 questions.

1. _____ , sql query
2. - - - -
⋮
50.

10 easy questions
20 med "
30 hard "



③ evaluation with golden dataset.

q1 → 5-6 Tera → SQL → DB run → llm query

golden DB query.

Result match:

~~vaide~~ Programmatically. Check output.

q2
⋮
950

True → 1
False → 0

Now calculate accuracy For each model:

~~Now~~

How to prepare golden Dataset

Human Made → require money.

LLM calls → require money.

☆

custom eval:

✓ 1) The dataset → [ipl → 2020 - 2024.]

↓

✓ i) CSV File → Fetch Data & Create DataBase#

ii) Filter & load the Dataset.

✓ 3) extract the schema: [schema_extractor.py.]

④ Build the golden Dataset → Made with the Human experts.

Make 20 queries:

[golden_dataset_generator.py.]

GPT 5-6 Terq ——— 80

Kimi K3 ——— 55

Grok 4-5 ——— 90

claude sonnet 5 ——— 85

Minimax M3. ——— 65